**Full Stack Assignment – 12**

**Assignment ID:** FSD-12
**Title:** Room Booking Management System
**Duration:** 3 Hours

## 1️⃣ Objective

Design and develop a **Room Booking Management System** that enables users to book rooms and administrators to manage room availability.
The system must prevent booking conflicts and ensure **data persistence using a database**.

## 2️⃣ User Roles

**USER**

- Register and log in to the system
- View available rooms
- Book a room
- View own booking history

**ADMIN**

- Log in to the system
- Create and manage rooms
- View all room bookings

## 3️⃣ Functional Requirements

**Authentication & Authorization**

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

**Room Management**

- ADMIN can create rooms with capacity/details
- Room data must be stored in the database
- Room availability must be fetched dynamically

**Room Booking**

- USER can book rooms for specific time slots
- Booking details must be stored in the database

- Booking status must be retrievable

## 4️⃣ Business Rule (Critical)

⚠️ **The same room must not be booked for overlapping time slots.**

Any overlapping booking attempts must be rejected at the backend.

## 5️⃣ Technical Constraints (Mandatory)

- All room and booking data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect real-time database state
- JWT authentication and role validation are mandatory
- Environment variables must be used for sensitive configuration

## 6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ADMIN
- Room listing with availability
- Room booking action for USER
- Booking overview for ADMIN

UI appearance is not evaluated; correctness and logic matter.

## 7️⃣ Submission Requirements (Strict)

**GitHub Repository**

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

**Live Deployment**

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

**README.md (Mandatory)**

The README file must include:

- Project overview
- Tech stack used

- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

### 8️⃣ Evaluation Note

Evaluation focuses on **conflict detection, role enforcement, and end-to-end execution**.